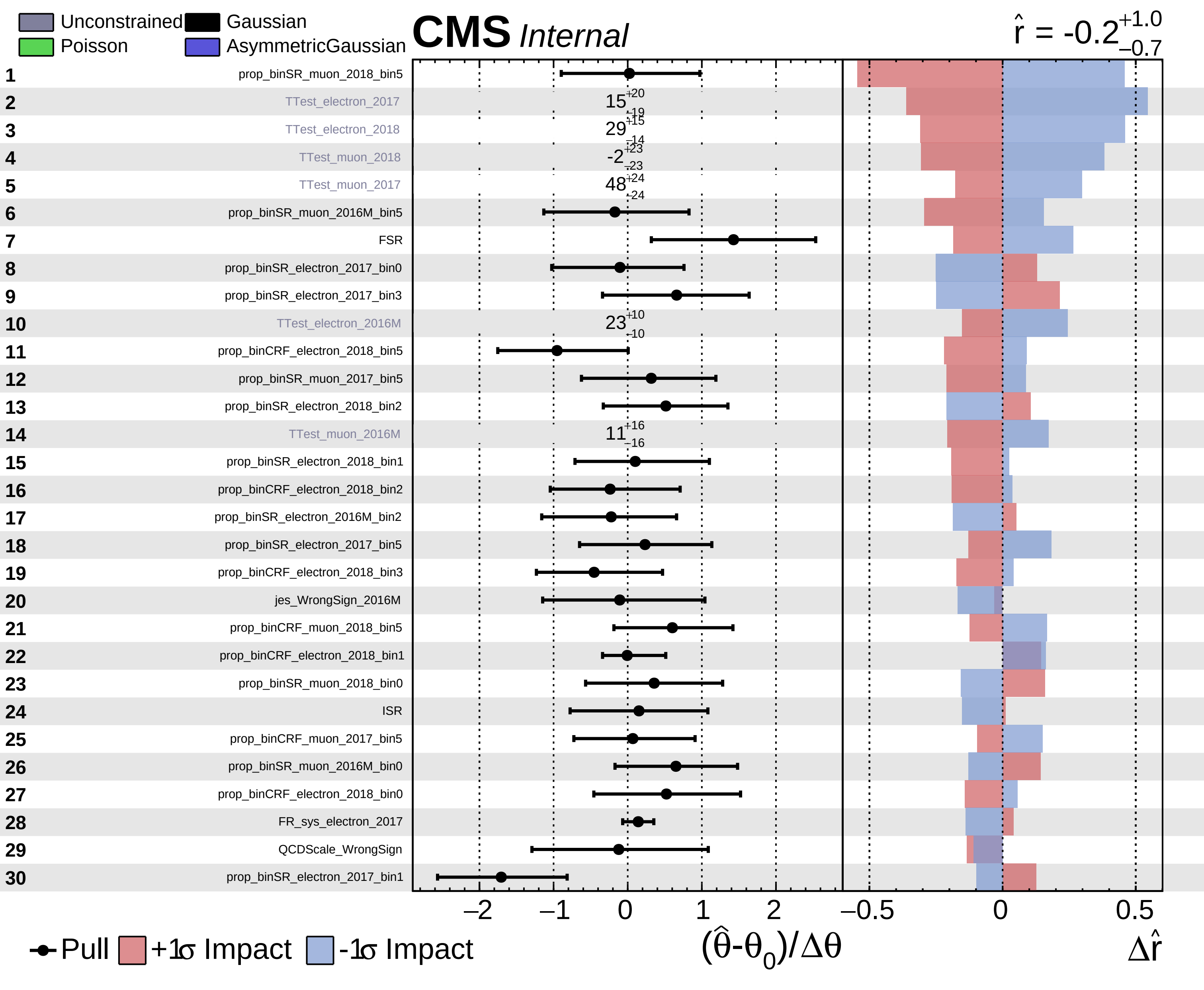
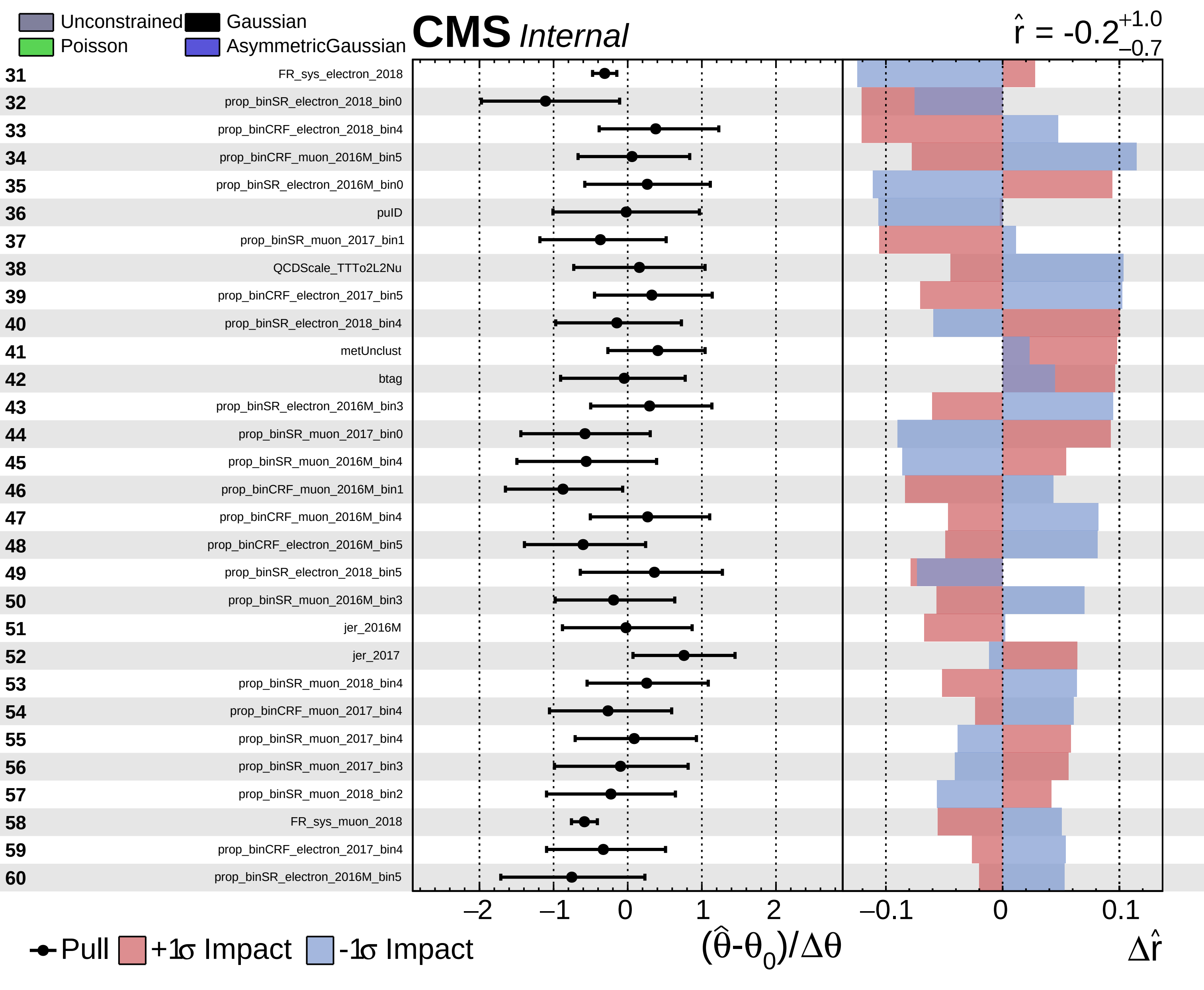
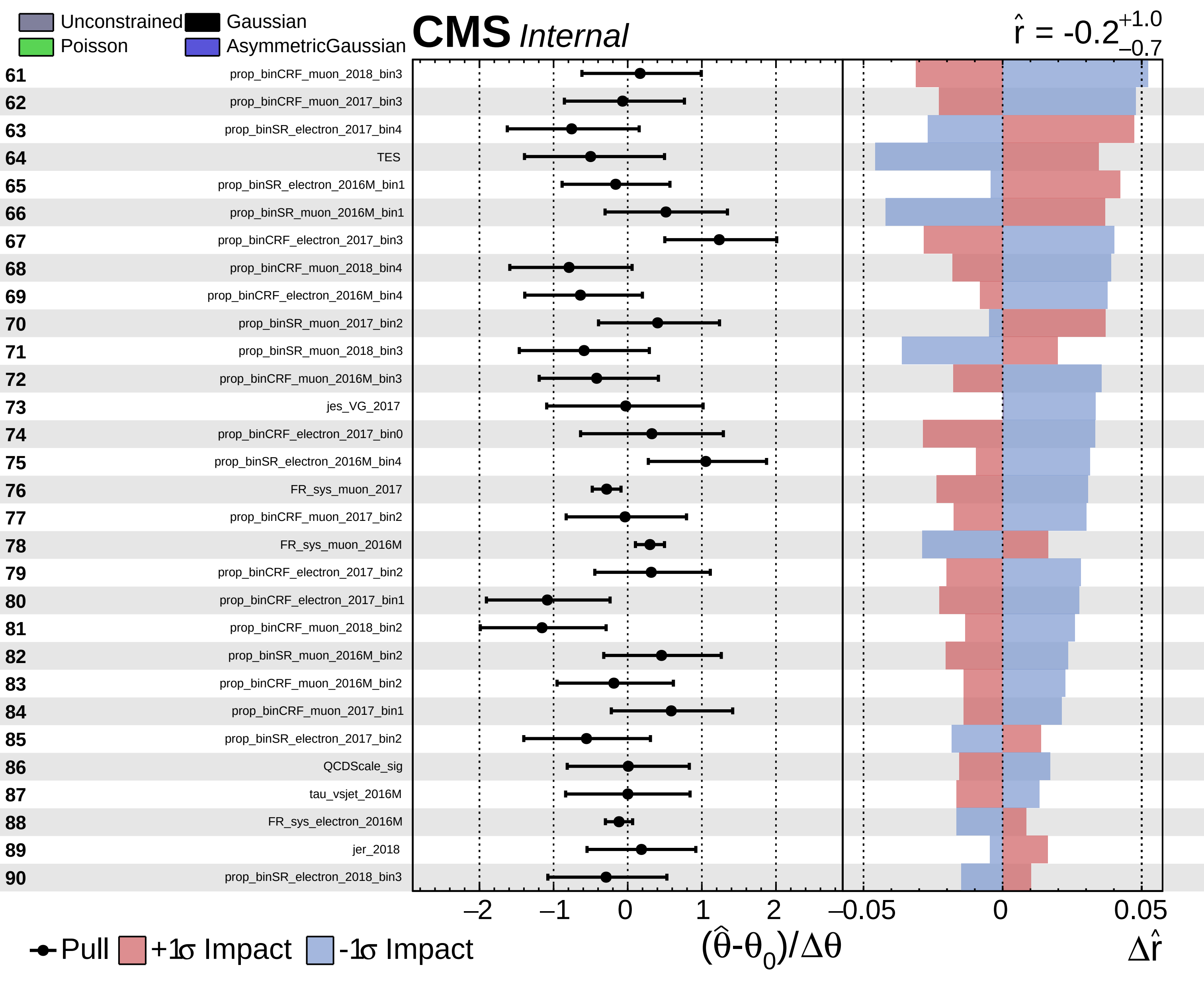


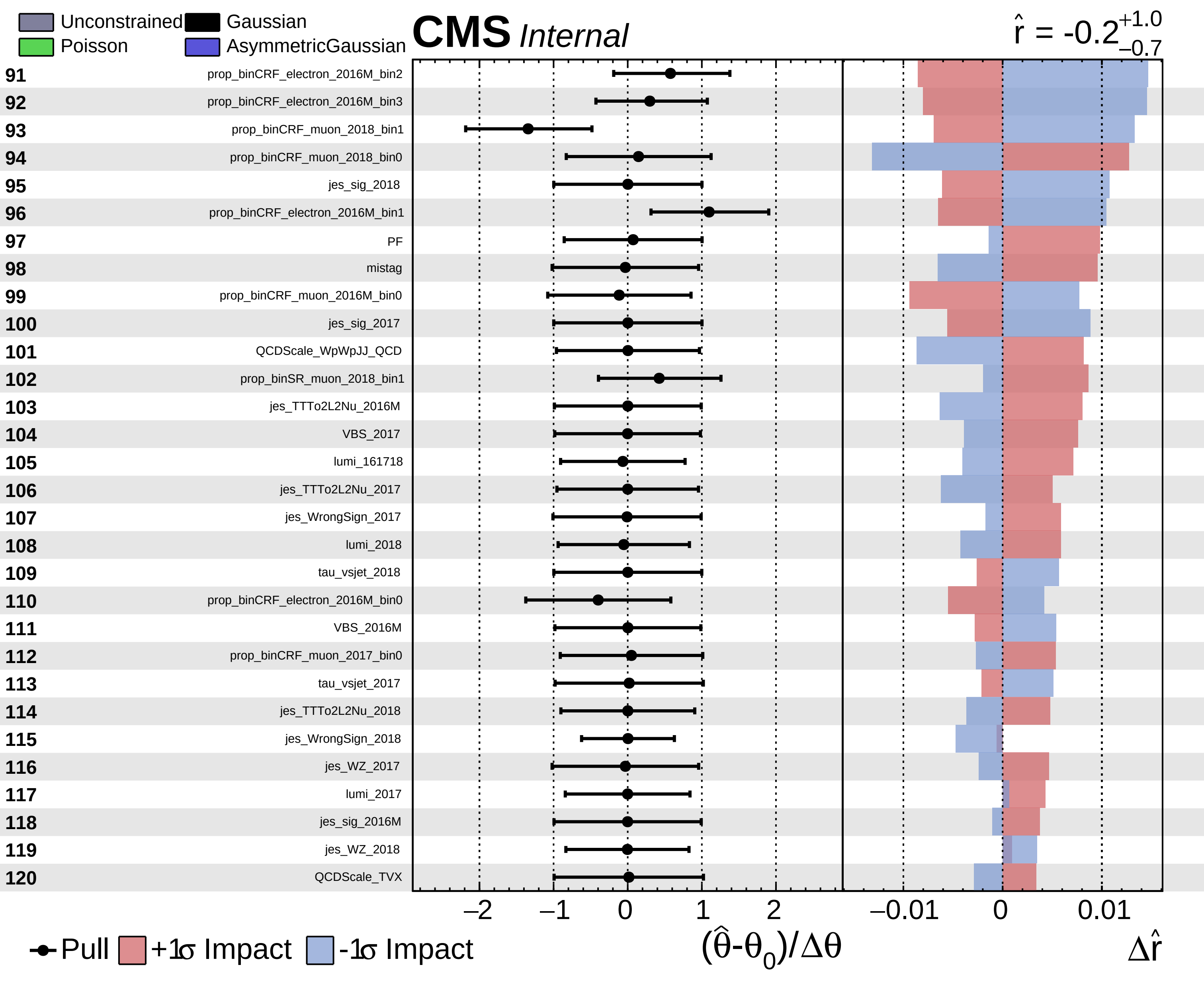
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.2^{+1.0}_{-0.7}$
1 prop_binSR_muon_2018_bin5
2 TTest_electron_2017 15^{+20}_{-19}
3 TTest_electron_2018 29^{+15}_{-14}
4 TTest_muon_2018 -2^{+23}_{-23}
5 TTest_muon_2017 48^{+24}_{-24}
6 prop_binSR_muon_2016M_bin5
7 FSR
8 prop_binSR_electron_2017_bin0
9 prop_binSR_electron_2017_bin3
10 TTest_electron_2016M 23^{+10}_{-10}
11 prop_binCRF_electron_2018_bin5
12 prop_binSR_muon_2017_bin5
13 prop_binSR_electron_2018_bin2
14 TTest_muon_2016M 11^{+16}_{-16}
15 prop_binSR_electron_2018_bin1
16 prop_binCRF_electron_2018_bin2
17 prop_binSR_electron_2016M_bin2
18 prop_binSR_electron_2017_bin5
19 prop_binCRF_electron_2018_bin3
20 jes_WrongSign_2016M
21 prop_binCRF_muon_2018_bin5
22 prop_binCRF_electron_2018_bin1
23 prop_binSR_muon_2018_bin0
24 ISR
25 prop_binCRF_muon_2017_bin5
26 prop_binSR_muon_2016M_bin0
27 prop_binCRF_electron_2018_bin0
28 FR_sys_electron_2017
29 QCDScale_WrongSign
30 prop_binSR_electron_2017_bin1
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

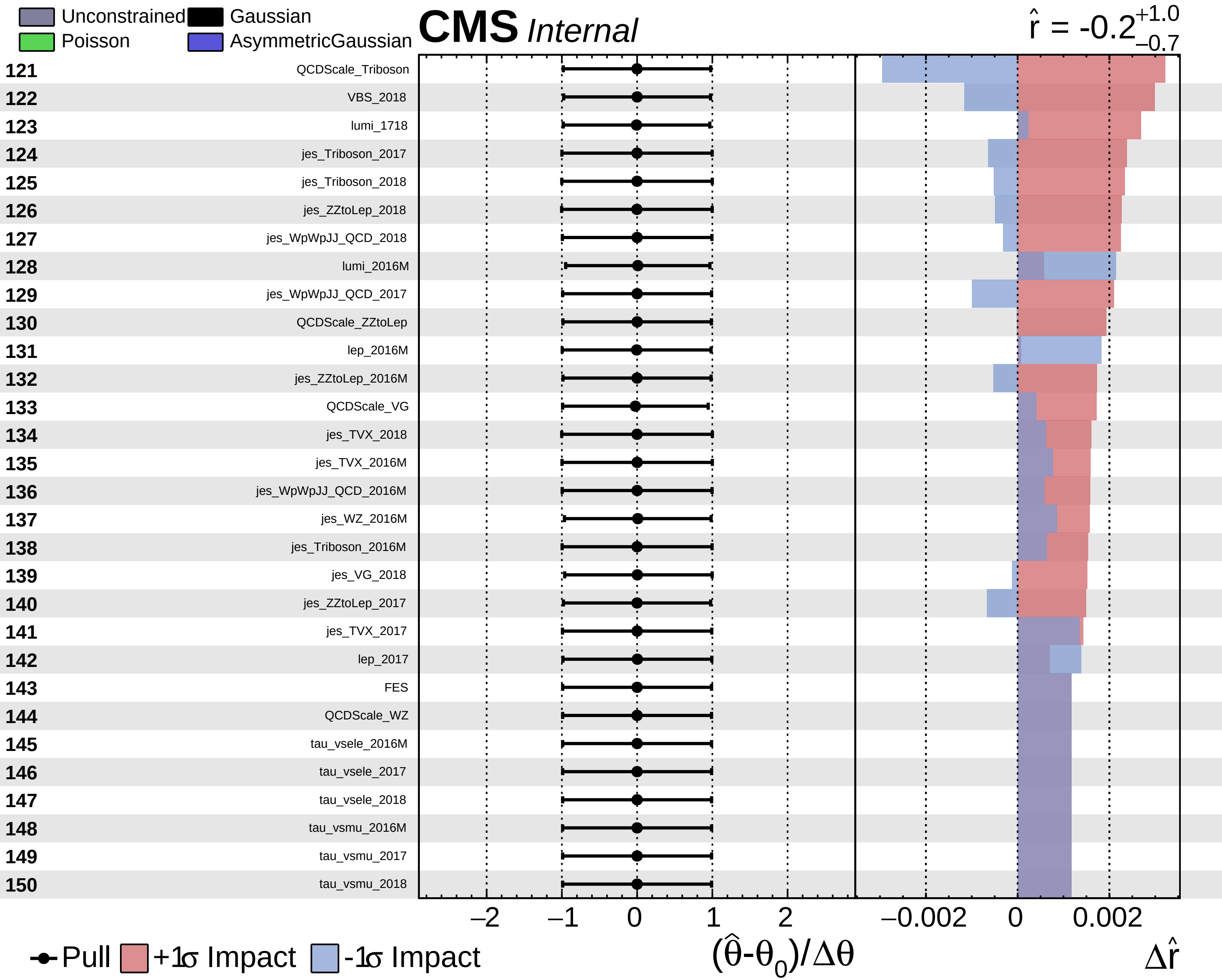


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r}$ = -0.2$^{+1.0}_{-0.7}$
31 FR_sys_electron_2018
32 prop_binSR_electron_2018_bin0
33 prop_binCRF_electron_2018_bin4
34 prop_binCRF_muon_2016M_bin5
35 prop_binSR_electron_2016M_bin0
36 puID
37 prop_binSR_muon_2017_bin1
38 QCDScale_TTTo2L2Nu
39 prop_binCRF_electron_2017_bin5
40 prop_binSR_electron_2018_bin4
41 metUnclust
42 btag
43 prop_binSR_electron_2016M_bin3
44 prop_binSR_muon_2017_bin0
45 prop_binSR_muon_2016M_bin4
46 prop_binCRF_muon_2016M_bin1
47 prop_binCRF_muon_2016M_bin4
48 prop_binCRF_electron_2016M_bin5
49 prop_binSR_electron_2018_bin5
50 prop_binSR_muon_2016M_bin3
51 jer_2016M
52 jer_2017
53 prop_binSR_muon_2018_bin4
54 prop_binCRF_muon_2017_bin4
55 prop_binSR_muon_2017_bin4
56 prop_binSR_muon_2017_bin3
57 prop_binSR_muon_2018_bin2
58 FR_sys_muon_2018
59 prop_binCRF_electron_2017_bin4
60 prop_binSR_electron_2016M_bin5
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

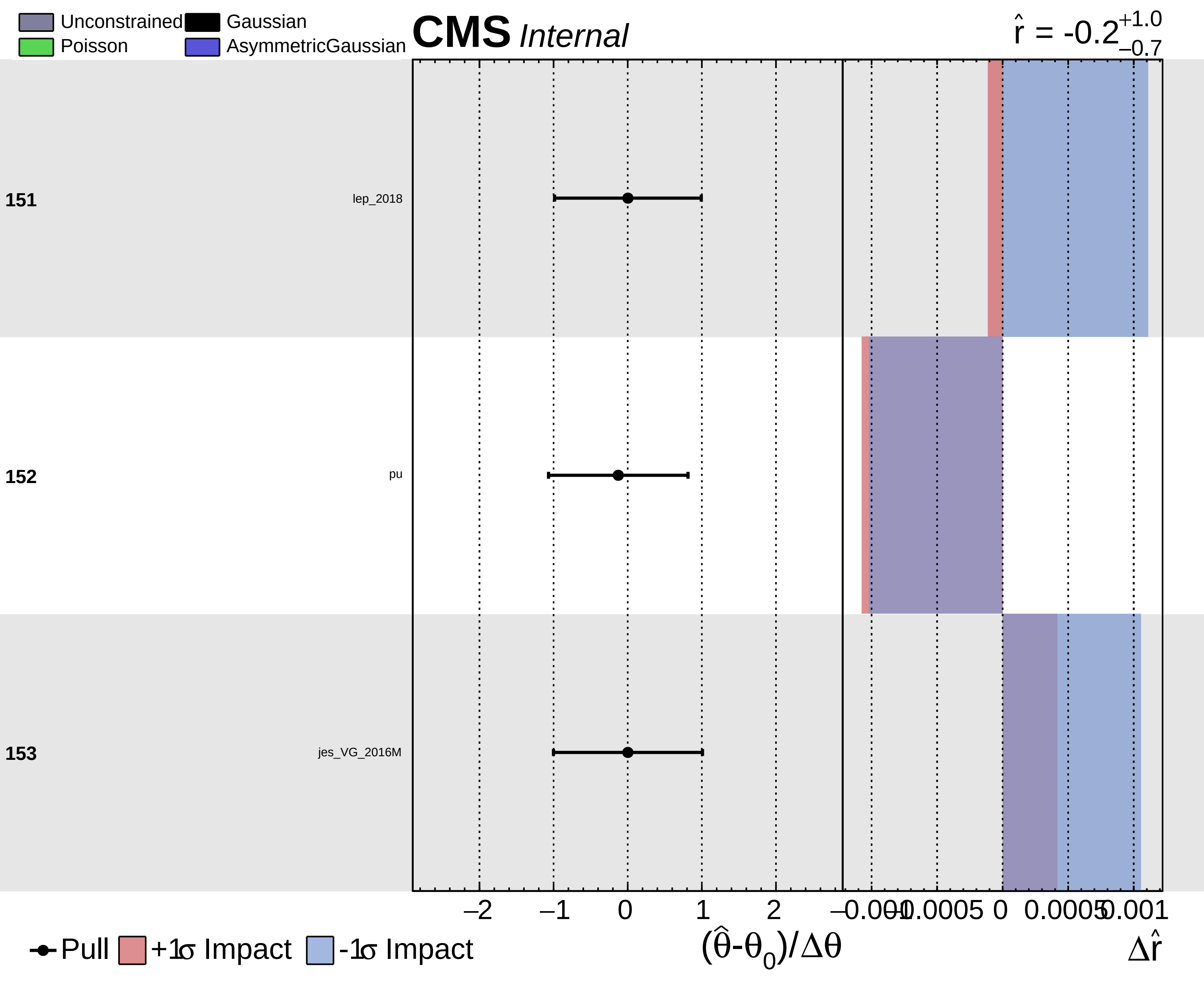


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.2^{+1.0}_{-0.7}$
61 prop_binCRF_muon_2018_bin3
62 prop_binCRF_muon_2017_bin3
63 prop_binSR_electron_2017_bin4
64 TES
65 prop_binSR_electron_2016M_bin1
66 prop_binSR_muon_2016M_bin1
67 prop_binCRF_electron_2017_bin3
68 prop_binCRF_muon_2018_bin4
69 prop_binCRF_electron_2016M_bin4
70 prop_binSR_muon_2017_bin2
71 prop_binSR_muon_2018_bin3
72 prop_binCRF_muon_2016M_bin3
73 jes_VG_2017
74 prop_binCRF_electron_2017_bin0
75 prop_binSR_electron_2016M_bin4
76 FR_sys_muon_2017
77 prop_binCRF_muon_2017_bin2
78 FR_sys_muon_2016M
79 prop_binCRF_electron_2017_bin2
80 prop_binCRF_electron_2017_bin1
81 prop_binCRF_muon_2018_bin2
82 prop_binSR_muon_2016M_bin2
83 prop_binCRF_muon_2016M_bin2
84 prop_binCRF_muon_2017_bin1
85 prop_binSR_electron_2017_bin2
86 QCDScale_sig
87 tau_vsjet_2016M
88 FR_sys_electron_2016M
89 jer_2018
90 prop_binSR_electron_2018_bin3
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.2^{+1.0}_{-0.7}$
91 prop_binCRF_electron_2016M_bin2
92 prop_binCRF_electron_2016M_bin3
93 prop_binCRF_muon_2018_bin1
94 prop_binCRF_muon_2018_bin0
95 jes_sig_2018
96 prop_binCRF_electron_2016M_bin1
97 PF
98 mistag
99 prop_binCRF_muon_2016M_bin0
100 jes_sig_2017
101 QCDScale_WpWpJJ_QCD
102 prop_binSR_muon_2018_bin1
103 jes_TTTo2L2Nu_2016M
104 VBS_2017
105 lumi_161718
106 jes_TTTo2L2Nu_2017
107 jes_WrongSign_2017
108 lumi_2018
109 tau_vsjet_2018
110 prop_binCRF_electron_2016M_bin0
111 VBS_2016M
112 prop_binCRF_muon_2017_bin0
113 tau_vsjet_2017
114 jes_TTTo2L2Nu_2018
115 jes_WrongSign_2018
116 jes_WZ_2017
117 lumi_2017
118 jes_sig_2016M
119 jes_WZ_2018
120 QCDScale_TVX
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.2^{+1.0}_{-0.7}$
121 QCDScale_Triboson
122 VBS_2018
123 lumi_1718
124 jes_Triboson_2017
125 jes_Triboson_2018
126 jes_ZZtoLep_2018
127 jes_WpWpJJ_QCD_2018
128 lumi_2016M
129 jes_WpWpJJ_QCD_2017
130 QCDScale_ZZtoLep
131 lep_2016M
132 jes_ZZtoLep_2016M
133 QCDScale_VG
134 jes_TVX_2018
135 jes_TVX_2016M
136 jes_WpWpJJ_QCD_2016M
137 jes_WZ_2016M
138 jes_Triboson_2016M
139 jes_VG_2018
140 jes_ZZtoLep_2017
141 jes_TVX_2017
142 lep_2017
143 FES
144 QCDScale_WZ
145 tau_vsele_2016M
146 tau_vsele_2017
147 tau_vsele_2018
148 tau_vsmu_2016M
149 tau_vsmu_2017
150 tau_vsmu_2018
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.2^{+1.0}_{-0.7}$
151
lep_2018
152
pu
153
jes_VG_2016M
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$